



Spirit and Opportunity

Learnings from Sojourner paved the way for the twin "geologist" rovers, Spirit and Opportunity. The objective of these rovers was to land on opposite sides of the Red Planet and explore the rocks and soil there for past water activity. Scientists used data from these rovers to reconstruct Mars' ancient history. Both these rovers explored far more surface than any rover before and returned more than 100,000 images before dying.

Spirit was launched on June 10, 2003, onboard the Delta 7925-9.5 rocket, and the rover landed on Mars on January 4, 2004, in the Gusev Crater. Opportunity was launched on July 8, 2003, onboard the Delta 7925H rocket, and it reached the Meridiani Planum landing site on Mars on January 25, 2004.

Though originally planned for 90-day missions, both Spirit and Opportunity lived well beyond their lifetimes. Spirit kept on roving from one crater to the other over the course of its lifetime — a natural cleaning of its solar panels thanks to dust storms on Mars further kept propelling the mission. Soon, it developed issues with its wheels and storms on the planet reduced the output of its solar panels. NASA finally ended contact with Spirit on May 25, 2011.

Opportunity had its share of firsts too. It was the first rover to exceed a marathon travel distance on another world. It also holds the record for ascending the steepest slope driven by any Martian rover. NASA extended Opportunity's missions several times, and finally declared its mission complete on February 13, 2019.

CURRENT ROVERS — PROBING FOR SIGNS OF MARTIAN LIFE

The first rovers gave us a wealth of

information about Mars' geological features and learnings from these rovers paved the way for successive, more sophisticated designs.

Curiosity

Curiosity launched on November 26, 2011, onboard the Atlas V-451 rocket and was destined for the Gale Crater on Mars on August 5, 2012. Curiosity, part of the Mars Science Laboratory mission, is one of the largest rovers on Mars with an array of advanced instrumentation including 10 science instruments, 17 cameras, a laser, and a drill.

So far, Curiosity has sent a total of 975,610 images and has spent ten years on the Martian surface, continuing to offer vital data. It also discovered fluctuating methane levels in the atmosphere, performed the first other-worldly rock dating, presence of jarosite, a molecule required for RNA, DNA, and protein synthesis. Radiation levels measured by Curiosity helped NASA determine that a manned mission to Mars is feasible in the future. The rover continues to explore Mars to this day, well past its two-year mission time.



Perseverance

Building on from Curiosity, NASA designed the Perseverance rover with a major aim of unlocking Mars' astrobiology. Part of the Mars 2020 Mission, the Perseverance rover seeks to study habitability that supports microbial life, collecting samples of the Martian surface, and laying the groundwork for future missions.

NASA launched Perseverance aboard the Atlas V-541 rocket on July 30, 2020. The rover landed at the Jezero Crater on February 18, 2021. Scientists suspect that the structure of the Jezero Crater makes it a likely ancient water reservoir.

The rover also carries with it a Mars Helicopter, which is a tech demo to understand powered flight in the Martian atmosphere. Learnings from this helicopter will let scientists determine the feasibility of designing robotic scouts in future missions for aerial surveillance.

Perseverance is designed to last for at least one Martian year (687 Earth days) and, if experience is any indication, the rover will outlast its shelf life for many more years to come.

Zhurong

China's Tianwen-1 mission is the country's first Mars mission with a lofty aim of putting both an orbiter and a rover on Mars in the very first attempt. Tianwen-1 launched on China's largest rocket, the Long March-5, on July 23, 2020. The orbiter entered the Martian orbit in February 2021 while the rover, named Zhurong, landed in the Utopia Planitia region on May 14, 2021.

Similar to NASA's Perseverance, Zhurong also aims to cache samples for eventual return to Earth. Zhurong has about 14 onboard instruments for collecting scientific data from the Martian atmosphere and rocks. It also carries with it a radar for underground 3D surface mapping to discover potential water ice.

Zhurong has an expected lifespan of at least 90 Martian days while the orbiter is slated to last for one Martian year (687 Earth days).

DESTINATION, MARS?

Rovers, past and present, have given equipped us with knowledge and opened new avenues hitherto thought to be impossible. There are certain ethical considerations as well. Such as protecting the Red Planet from what is known as forward contamination. Forward contamination refers to transfer of microorganisms from spacecraft onto Mars. If life exists of Mars, the contamination from spacecraft and future humans can override the natural life in that area leading to errors in scientific experiments. Such questions are somethings we need to consider for sure as we continue to explore. 